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Low-noise circuitry for low-temperature NMR and SQUIDS

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A method of coupling the low-temperature induction coil of an NMR or SQUID system to a preamplifier is proposed. This method greatly reduces the Johnson noise generated by line losses. A suitable 9-K noise-temperature GaAs FET, liquid nitrogen cooled preamplifier is described. The coupling scheme and preamplifier will yield a spectrometer with a noise temperature near 15 K.

Low-temperature NMR experiments and rf-biased SQUID detection schemes often require the amplification of small rf signals. Because the signal source is at a low temperature, the detection apparatus may in principle have a low noise figure and be capable of detecting correspondingly weak signals. However, this goal usually has not been achieved in practice. Some satisfactory techniques are described below.

The most common approach to coupling the low-temperature coil to a preamplifier is shown in Fig. 1. The transmission line is part of the resonant circuit and carries the circulating current that flows through L , C , and C' , the distributed capacitance of the line. Alderman¹ pointed out that much of the loss in this tuned circuit occurs in the transmission line near room temperature. Hence, the noise temperature of the tuned circuit is about 200 K. This may be improved by placing C at the low-temperature end of the line and by keeping C' small, causing less circulating current to flow through the line. In addition, use of a large C and small L reduces the noise temperature of the circuit. One can estimate a noise temperature using these techniques for frequencies of 10 MHz and higher to be of order 100 K, with practical circuit values. At much lower frequencies these methods perform better.²

Alderman¹ and Miyoshi and Cotts³ placed preamplifiers employing silicon MOSFETs in the liquid helium experimental bath. This reduces the input-circuit noise temperature to the physical temperature of the coil. However, the MOSFETs are fairly noisy at this temperature; Miyoshi and Cotts³ reported an 88-K noise temperature for their preamplifier. Better performance can be attained with cooled GaAs FETs. Measurements at 4.2 K reported below show a 28-K noise temperature. However, location of the amplifier in the helium bath

often requires a crossed-coil design,³ a bridge at the low temperature,¹ or (in pulsed work) a low-temperature transmit-receive switching and protection scheme. In crowded experimental probes this is inconvenient. Further, the GaAs FET dissipates typically 50 mW into the liquid helium, increasing the boil-off rate.

A simple alternative, shown in Fig. 2, is to impedance match the low-temperature tuned circuit to a transmission line. A convenient method is to "tap down" on the tuned circuit. Because the line is matched to the tuned circuit, the line losses are small and consequently generate little Johnson noise (each 0.1 dB loss generates about 6.9 K of noise, for losses under 0.5 dB). Hence, the input-circuit noise temperature is essentially the physical temperature of the coil. The low line losses make possible a very high- Q tuned circuit at the low temperature, which also enhances sensitivity. Although this matched-line approach is used in vhf NMR, its advantages for low-temperature operation at all frequencies apparently have not been recognized.

When the input noise is so reduced, the noise of conventional preamplifiers becomes quite significant. Hoult and Richards⁴ have constructed a liquid nitrogen cooled, GaAs FET preamplifier with a 21.5-K noise temperature. We have constructed a GaAs FET preamplifier for 20 MHz which has a 9-K noise temperature when cooled by liquid nitrogen. Use of a low-temperature NMR or SQUID tuned input circuit, a matched line, and a nitrogen cooled, GaAs FET preamplifier will yield at 20 MHz an overall noise temperature near 15 K (4-K coil, 2-K line loss, 9-K preamplifier), considerably lower than those already reported. Additionally, this scheme is easy to incorporate into crowded experimental probes and dissipates no power into the liquid helium bath.

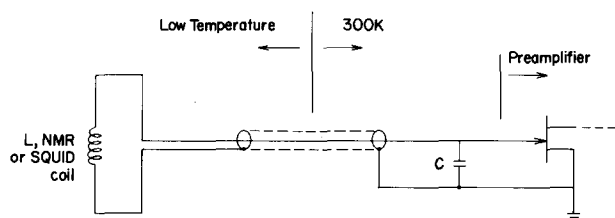


FIG. 1. Usual method of connecting induction coil L to the preamplifier.

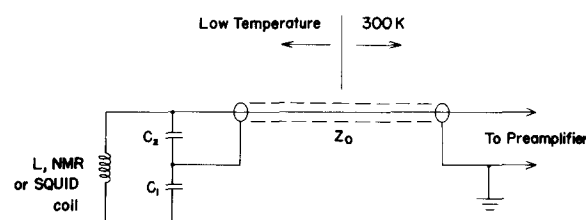


FIG. 2. Matched-line approach to coupling signal to preamplifier. L resonates C_1 and C_2 in series. Matching condition is line impedance $= Z_0 = Q\omega L[C_1/(C_1 + C_2)]^2$, where Q is the unloaded Q of L .

The differences between the 20-MHz preamplifier and that of Hoult and Richards⁴ warrant a brief description of the former. As shown in Fig. 3, this is a cascode design with only the first stage cooled. The resonant input-matching circuit transforms a 93- Ω input source to a 2000- Ω source at the FET gate. Cooling with liquid nitrogen reduces the Johnson noise power generated by this matching circuit by a factor of 10: the temperature is reduced from room temperature by a factor of 4 and the unloaded Q of the matching coil is increased a factor of 2.5. The matching circuit contributes about 2 K to the overall noise temperature.

The GaAs FET, a Plessey Semiconductors GAT-1/010, is mounted in the liquid with the top of its can removed, ensuring that the active element is cold. These devices are intended for 1-GHz service and easily generate parasitic uhf oscillations. Components R_1 , R_2 , and C_1 prevent these oscillations. A pencil or screwdriver touched to one of the leads of R_2 will produce a small change in amplifier dc current if uhf oscillation is present.

The first stage is built in a copper can of 6-cm inner diameter and 10-cm length; liquid nitrogen surrounds and fills this can. The can is made large to allow use of a large and high- Q input-matching coil. The can is supported by the 16-cm-long outer conductors of the two coaxial lines, which are thin-wall stainless steel. The second stage is of conventional design and can be shown to contribute little noise.

The noise temperature of the preamplifier was measured to be 9 K using the hot-cold source technique. The source was placed at 300, 77, and 4.2 K (while the preamplifier temperature was unchanged). At each source temperature the noise power at the output was measured in a narrow band at 20 MHz. A second and similar amplifier, intended for liquid helium immersion, was built with a lower Q input-matching coil and a longer line connecting the two cascode halves. With this amplifier at 300, 77, and 4.2 K the noise temperatures were 43, 15, and 28 K, respectively. The increased noise at 4.2 K

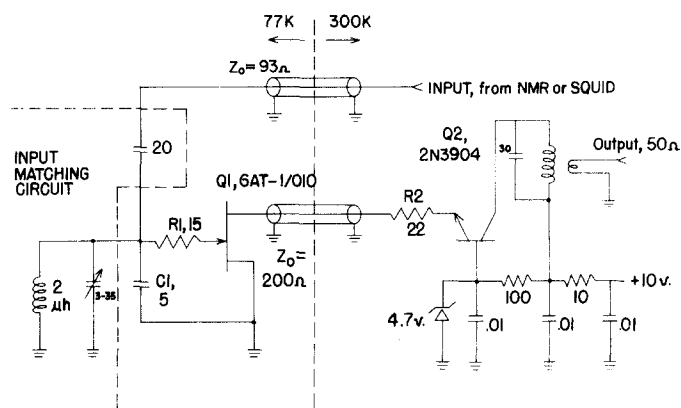


FIG. 3. Circuit of the 77-K GaAs FET amplifier for 20 MHz. Resistances are in ohms, fractional capacitance values are in μF , integer capacitance values are in pF.

is not understood at present. Finally, we note some results with GaAs FETs at liquid hydrogen temperature and 5 GHz.⁵

With the input port loaded, signal-frequency oscillations did not occur, reflecting the inherent stability of the cascode design. The cascode is simple to build and tune and has sufficient gain to mask the noise of subsequent stages.

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⁵ J. Pierro; see *Microwaves* **15**, 10 (June 1976).

Piston gauge weight handler

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An electromechanical hydraulic lift has been developed and subsequently used to load and unload weight stacks from precision piston gauges without affecting the reliability or the operation of these gauges.

In an extensive gas PVT program two controlled-clearance piston gauges, each for a different pressure range, are being used to measure indirectly the pressure of the gas being studied. This gas is separated by a sensitive differential pressure transducer from pres-

sure measuring instrumentation. The pressure measured is that of the transducer counterbalancing gas used to null the transducer. It corresponds as well to the fluid pressure generated by one of the piston gauges loaded with the appropriate number of weights. The pressure is